

Grade 11 Chemistry November Memo 2021

Question 1

20 Marks

1.1	B	✓	✓
1.2	B	✓	✓
1.3	B	✓	✓
1.4	D	✓	✓
1.5	A	✓	✓
1.6	D	✓	✓
1.7	C	✓	✓
1.8	D	✓	✓
1.9	C	✓	✓
1.10	D	✓	✓

Question 2 Bonding

23 Marks

2.1.1 Solid Li_2O has **no free ions** ✓ to act as current carriers because the ions are **fixed in the crystal lattice.** ✓ (2)

2.1.2 a. oxidation ✓✓ (1)

2.1.2 b. $\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$ ✓✓ (2)

2.1.3 covalent bond ✓ (1)

2.1.4 SiO_2 is a **giant covalent lattice** ✓
 SiO_2 has **many strong covalent bonds** ✓
More energy is needed to break these bonds than the ionic bonds in SnO_2 . ✓ (3)

2.2.1 A bond which occurs between atoms within molecules. ✓✓ (2)

2.2.2

2,1	3,0	
H	Cl	$\Delta\text{EN} = 3,0 - 2,1 = 0,9$ ✓
		polar covalent bond ✓

(2)

2.2.3 Hydrogen bond force ✓ (1)

2.2.4 **small highly electronegative** ✓ atom attracts the electron bonding pair
A **naked proton** is exposed ✓ and is attracted to a
Lone pair ✓ of electrons on another small highly electronegative atom (3)

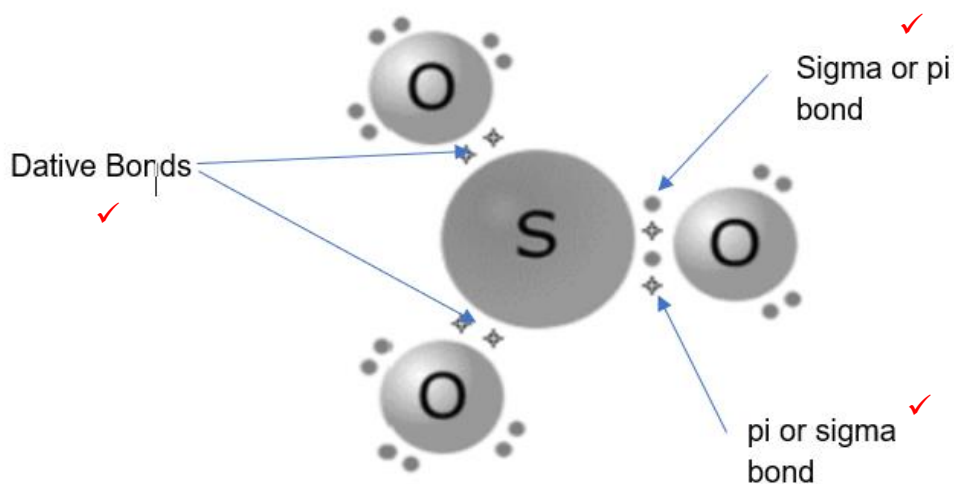
2.2.5

2.2.7 HBr and HI have **dipole-dipole forces** and **London Forces** ✓
HI is a **larger molecule** OR has **more electrons** ✓ and forms a larger temporary dipole
More energy will be needed to boil HI than HBr. HI has the higher boiling point
HI has the higher boiling point ✓ (3)

Question 3 Bonding

12 Marks

- 3.1.1 Bonding between a positive kernel ✓ and a sea of delocalised electrons. ✓ (2)
- 3.1.2 Refers to the nucleus ✓ and core electrons. ✓ (2)
- 3.1.3 delocalised electrons ✓ (1)
- 3.1.4 Mg has a bigger charge (+2) ✓
Greater attraction between the nucleus and the electron cloud. ✓ (2)
- 3.2.1 empty orbital ✓
Lone pair of electrons ✓ (2)
- 3.2.2 ANSWER SHEET (3)



Question 4**25 Marks**

- 4.1 Potential Energy ✓ (kJ) ✓ (2)
- 4.2 ANSWER SHEET (5)

QUESTION 4.2	
I	ENERGY OF REACTANTS ✓
II	ACTIVATION ENERGY ✓
III	ENERGY RELEASED ✓
IV	ENERGY OF PRODUCTS ✓
V	HEAT OF REACTION ✓

- 4.3 Only particles with sufficient kinetic energy ($E_k \geq E_a$) ✓ equal to that of position B will be effective. ✓ (2)
- 4.4 Activated Complex ✓ Old bonds are broken here and new bonds are formed. ✓ (2)
- 4.5 Exothermic ✓ More thermal energy ✓ is released than is absorbed. ✓ or the heat of the reaction is negative OR E_p greater than E_r (3)
- 4.6.1 Increases the rate of a reaction ✓ but remains unchanged at the end of the reaction. ✓ (2)
- 4.6.2 DECREASES. ✓✓ (2)
- 4.6.3 Stability increases ✓
Less energy is needed to start the reaction ✓
Less energy implies more stability ✓ (3)
- 4.7 At C1 there are **more particles per unit volume** than at C2 ✓
There will be **more collisions per unit time** at C1 ✓
More **effective collisions** per unit time occurs ✓
 $E_k > \text{than } E_a$ which is the shaded area, rate increases ✓ (4)

MUST RELATE EXPLANATION TO GRAPH. IF NO REFERENCE TO GRAPH **2 OUT OF 4**

Question 5

17 Marks

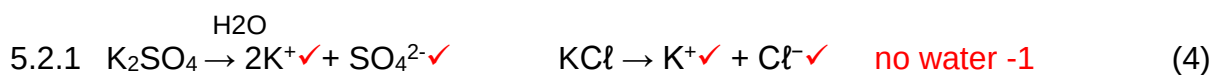
5.1.1 Solution of known concentration. ✓✓ (2)

5.1.2 $C = \frac{m}{MV}$ ✓

$$0,5 = \frac{m}{0,5 \times 174}$$

$$m = 43,5 \text{ g} \quad (4)$$

5.1.3 Before ✓ To ensure that the final volume is exactly 500 cm³ and not more than 500 cm³. ✓ (2)



5.2.2 **K₂SO₄**

$$n = cv$$

$$= 0,8 \times 0,6$$

$$= 0,48 \text{ mol} \quad \checkmark$$

KCl

$$n = 0,25 \times 0,3$$

$$= 0,075 \text{ mol} \quad \checkmark$$

$$[\text{K}^+] = 2 \times 0,48$$

$$= 0,96 \text{ mol} \quad \checkmark$$

$$C_{\text{new}} = \frac{0,96 + 0,075}{0,6 + 0,25} \quad \checkmark$$

$$= 1,22 \text{ mol.dm}^{-3} \quad \checkmark \quad (5)$$

Question 6**18 Marks**

6.1 Ryan is correct. ✓

1 mol of any gas at STP occupies a volume of 22,4 dm³ ✓Amount of substance can be determined by $n = \frac{V}{V_m}$ ✓If volume changes then the amount of substance also changes
thus n will be bigger for CO₂ ✓6.2.1 BaCl₂AgNO₃

$$n = \frac{1,39}{208,3}$$

$$= 0,0067 \text{ mol} \quad \checkmark$$

Ratio

BaCl₂ : AgNO₃

1 : 2 ✓

0,0067 : 0,0134

$$n = \frac{2,69}{170}$$

$$= 0,0158 \text{ mol} \quad \checkmark$$

There is excess AgNO₃, hence BaCl₂ is the limiting reactant. ✓ (4)

6.2.2 $m(\text{AgCl}) = 0,0134(108 + 35,5) \quad \checkmark$

$$= 1,92 \text{ g} \quad \checkmark \quad (2)$$

6.2.3 $\% \text{ yield} = \frac{\text{theoretical}}{\text{actual}} \times 100 \quad \checkmark$

$$= \frac{1,8 \quad \checkmark}{1,92 \quad \checkmark} \times 100$$

$$= 93,99 \%$$

$$= 94 \% \quad \checkmark \quad (4)$$

6.3.1 0,01 ✓

6.3.2 0,07 ✓

6.3.3 1:7 ✓

6.3.4 MgSO₄ · 7 H₂O ✓ (4)

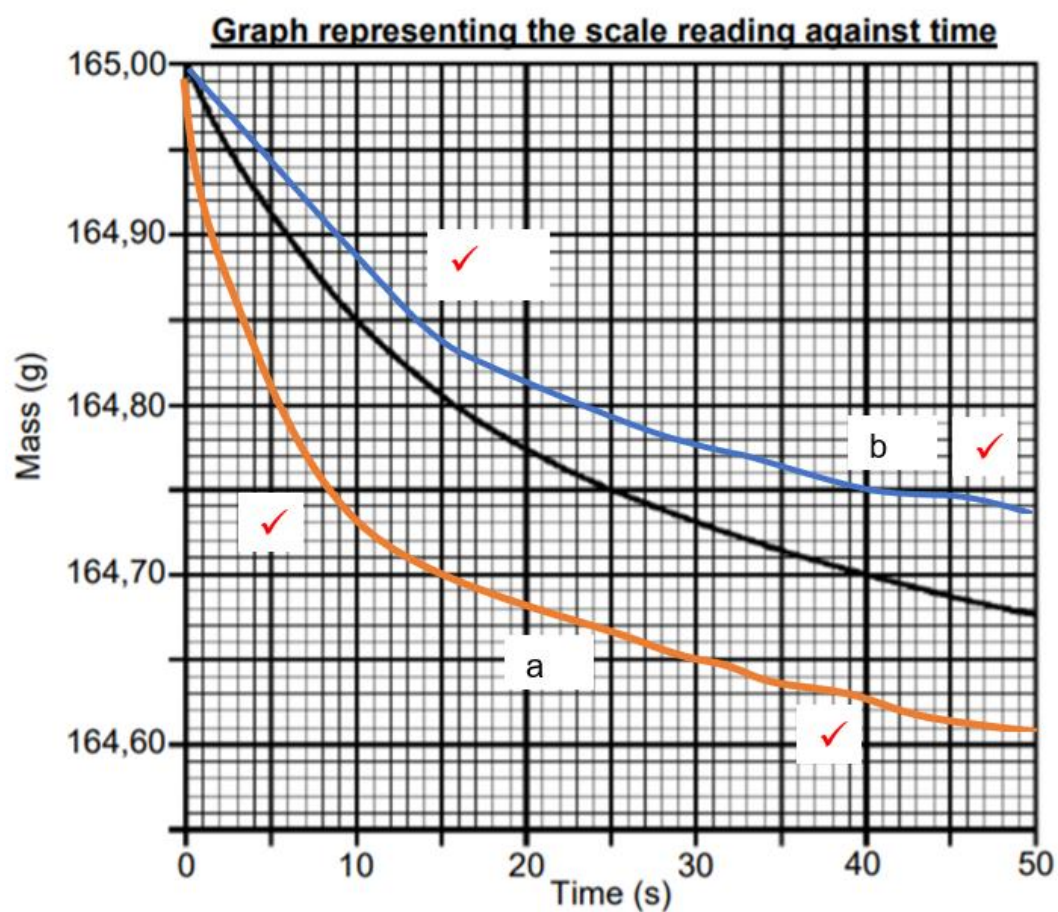
Question 7**29 Marks**

- 7.1 To ensure that only CO₂ escapes ✓ and that the rate of reaction is only measured by the decrease in concentration of CO₂. ✓. (2)
- 7.2 Change in concentration per unit time ✓ of either a reactant or product. ✓ (2)
- 7.3 No ✓ the graph has not yet levelled off. ✓ (2)
- 7.4 Use a syringe to collect the gas ✓ to get the volume produced per unit time. ✓ or the change in colour ✓ over a period of time can be measured ✓ (2)

7.5 average rate = $\frac{\Delta mass}{\Delta time}$
= $\frac{164,73 - 164,25}{30 - 10}$ ✓
= **0,006 g.s⁻¹**
Rate = $\frac{0,006}{44}$ ✓
= **1,36 x 10⁻⁴ mol.s⁻¹** ✓ (4)

7.6 a) & b

(4)



$$7.7.1 \quad n(\text{CO}_2) = \frac{v}{v_m} \checkmark = \frac{0,18}{22,4} = \mathbf{0,008 \text{ mol}} \checkmark$$

$$n(\text{CuCO}_3) = 0,008 \text{ mol} \checkmark$$

$$m(\text{CuCO}_3) = 0,008 \times 123,5 \\ = 0,99 \text{ g} \checkmark$$

$$\% \text{ purity} = \frac{0,99}{1,5} \checkmark \times 100 \\ = \mathbf{66 \%} \checkmark$$

(6)

7.7.2 amount of solute $\checkmark$ per unit volume of solution. $\checkmark$

(2)

$$7.7.3 \quad n(\text{HCl}) = 0,0075 \times 0,25 \\ = \mathbf{0,01875 \text{ mol}} \checkmark$$

$$\text{Amt of HCl left over} = 0,01875 - (0,008 \times 2) \checkmark \\ = \mathbf{0,00275 \text{ mol}} \checkmark$$

$$\begin{aligned}
 C &= \frac{0,00075}{0,075} \checkmark \\
 &= 0,037 \text{ mol.dm}^{-3} \checkmark
 \end{aligned}
 \tag{5}$$

Question 8**24 Marks**

8.1 B penta- ✓ 1,2-diene ✓ (2)

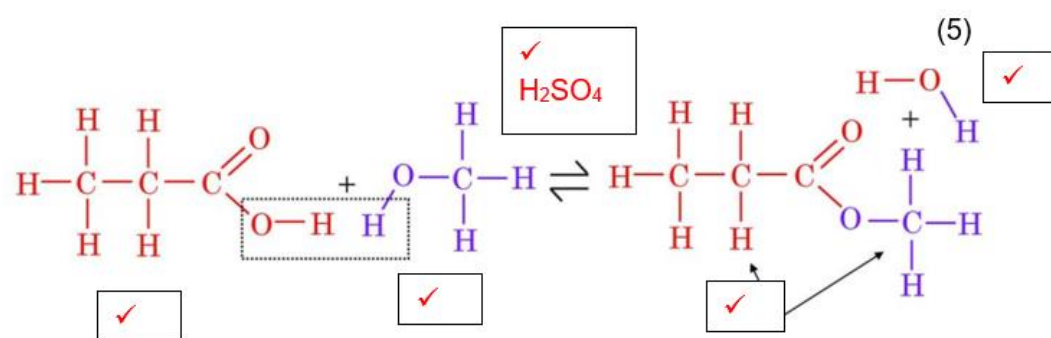
E propane- ✓ 1,3 -diol ✓ (2)

8.2.1 Compounds with the same molecular formula ✓ but a different structural formula. ✓ (2)

8.2.2 C ✓ (1)

8.2.3 C has a larger surface area ✓ and stronger London forces. ✓ (2)

8.4.1



8.4.2 butanoic acid ✓ $C_4H_8O_2$ ✓ (2)

8.5. Series of similar compounds with have the same functional group and the same general formula ✓ in which each member differs from the previous one by a single CH_2 unit. ✓ (2)

8.6 carboxylic acids ✓ (1)

8.7 C_nH_{2n+2} ✓ (1)

Question 9**32 Marks**

9.1.1 elimination ✓ (1)

9.1.2 thermal cracking ✓ (1)

9.2 (3)



1 reactants ✓

2 products ✓

3 balanced ✓

9.3.1 2- ✓ methyl ✓ pentane ✓ OR
2,3- ✓ dimethyl ✓ butane (3)9.3.2 $\text{CH}_3\text{CH}_2(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$ ✓ ✓ (2)

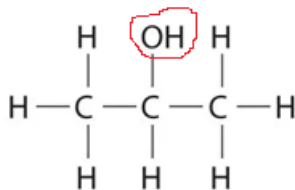
9.4 propene ✓ ✓ (2)

9.5.1 hydrohalogenation ✓ ✓ (2)

9.5.2 no water ✓ ✓ (2)

9.6.1 substitution ✓ ✓ (2)

9.6.2



1- Structure ✓

2- Circling OH ✓

3- Hydroxyl ✓

(3)

9.6.3 iv NaOH OR KOH ✓ (1)

v NaCl OR KCl ✓ (1)

9.6.4 heat under reflux, ✓ dilute alkali ✓ (2)

9.7.1 L – London forces ✓ (1)

M – dipole-dipole forces ✓ (1)

N – hydrogen bond force ✓ (1)

9.7.2 L, M, N ✓ ✓ (2)

9.7.3 N has the strongest force ✓ so more energy is needed to overcome
the forces of attraction for boiling to occur. ✓ (2)